

Progress Report

I. Project scope update

The project goal became more focused as I worked through the data. Originally, I wanted to explore PCOS using medical data, Reddit discussions, lifestyle factors, and Google Trends. However, Google Trends caused too many technical issues and did not connect clearly enough to the main question. Because of that, I narrowed the project to focus on whether observable PCOS symptoms in medical data match the symptoms people discuss in r/PCOS. The final project now compares medical symptom patterns with patient-reported symptom discussions. I also added a smaller lifestyle section to provide context around factors like stress, sleep, exercise, sweets intake, and fried food intake. This helped keep the project focused while still connecting medical symptoms, patient experience, and lifestyle behavior.

II. Data sources

I used three main data sources for this project. The first source is a PCOS medical dataset from Kaggle, which includes diagnosed and non-diagnosed PCOS patients. From this dataset, I focused on observable symptoms such as menstrual irregularity, cycle length, weight gain, hair growth, skin darkening, hair loss, and pimples. This source helped me calculate symptom prevalence and train classifiers to predict PCOS using observable medical symptoms.

The second source is r/PCOS Reddit data, which I collected using Python's requests library and Reddit's public JSON endpoint. The API request pulled recent posts from <https://www.reddit.com/r/PCOS/new.json>, and the returned JSON data included post titles, post body text, Reddit score, number of comments, date posted, and post permalink. I used keyword matching to identify how often observable symptoms were discussed in Reddit posts. This helped compare medical symptom data with patient-reported experiences.

The third source is a PCOS lifestyle dataset from Kaggle. I used this dataset to analyze lifestyle behavior factors such as stress level, sleep hours, exercise frequency, sweets intake, and fried food intake. This source helped provide additional context for how lifestyle factors may relate to PCOS experiences and management.

III. Issues / difficulties

One major challenge was narrowing the project scope. PCOS has many possible medical, lifestyle, and emotional factors, so the project became too broad at first. I had to focus specifically on observable symptoms that could be compared between the medical dataset and Reddit posts.

Another issue was Google Trends. I originally wanted to include it, but I ran into too many errors and it did not fit the final direction of the project well enough. I decided to remove it so I could focus on stronger and more relevant data sources.

Reddit data was also challenging because patient language is messy and unstructured. People describe symptoms in many different ways, so I had to create keyword categories and make sure they matched the medical symptom categories as closely as possible. I also had to remove lifestyle behaviors like fast food from the symptom comparison because they are not observable symptoms.

A final challenge was debugging the project structure. Connecting config.py, multiple Python files, dataset paths, chart outputs, and classifier scripts took a lot of time. A lot of the work involved fixing imports, matching column names, cleaning labels, and making sure the charts and models were consistent.